

Parametric models and logistic regression

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Warmup activity

Work on the activity (handout) with a neighbor, then we will discuss as a class.

Warmup

$$\text{odds} = \frac{\pi}{1 - \pi}$$

If $\pi = 0.2$, calculate the odds.

$$\text{odds} = \frac{0.2}{1 - 0.2} = 0.25$$

odds are not probabilities

Ex : if Age = S,
$$\frac{P(\text{Dengue} | \text{Age} = S)}{P(\text{no dengue} | \text{Age} = S)} = \frac{0.2}{1 - 0.2}$$

$$\Rightarrow P(\text{Dengue} | \text{Age} = S) = 0.25 P(\text{no dengue} | \text{Age} = S) = 0.25$$

Warmup

$$\text{odds} = \frac{\pi}{1 - \pi}$$

What happens to odds as $\pi \rightarrow 0$? As $\pi \rightarrow 1$?

$$\text{As } \pi \rightarrow 0, \quad \text{odds} \rightarrow 0$$

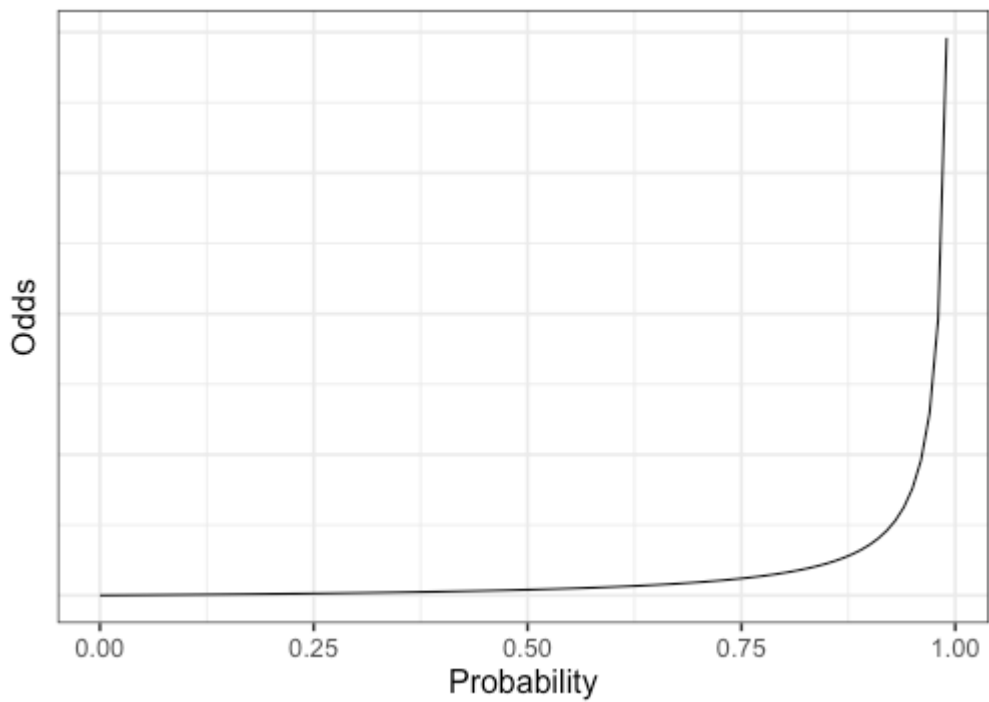
$$\text{As } \pi \rightarrow 1, \quad \text{odds} \rightarrow \infty$$

$$\Rightarrow \pi \in [0, 1]$$

$$\text{odds} \in [0, \infty)$$

Warmup

$$\text{odds} = \frac{\pi}{1 - \pi}$$



Warmup

$$\text{log-odds} = \log(\text{odds}) = \log\left(\frac{\pi}{1-\pi}\right)$$

What happens to log-odds as $\pi \rightarrow 0$? As $\pi \rightarrow 1$?

as $\pi \rightarrow 0$, odds $\rightarrow 0$, log-odds $\rightarrow -\infty$

as $\pi \rightarrow 1$, odds $\rightarrow \infty$, log-odds $\rightarrow \infty$

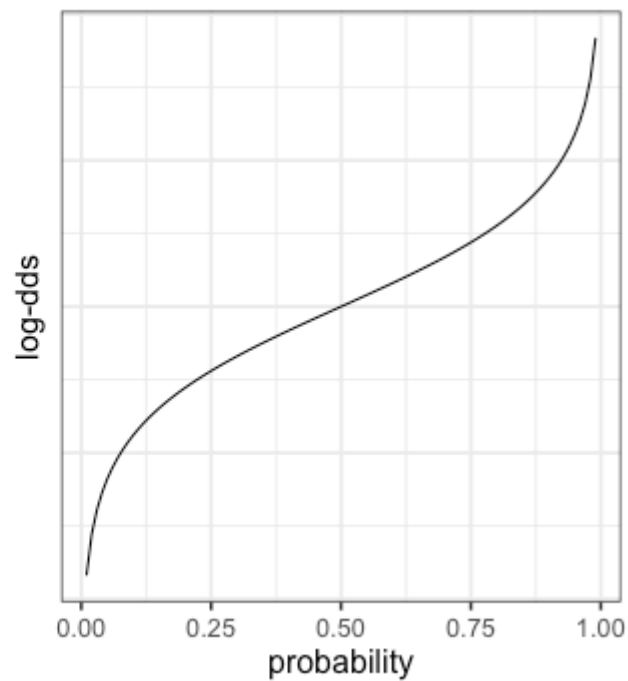
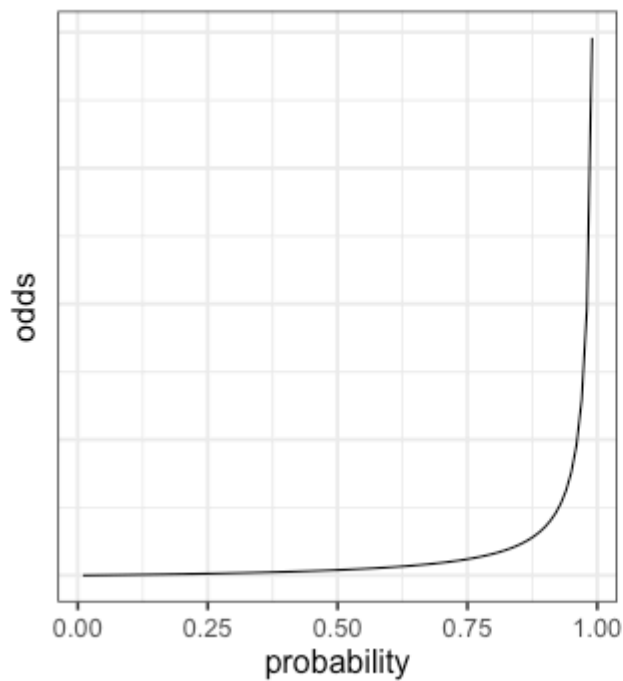
$$\pi \in [0, 1]$$

$$\text{odds} \in [0, \infty)$$

$$\text{log-odds} \in (-\infty, \infty)$$

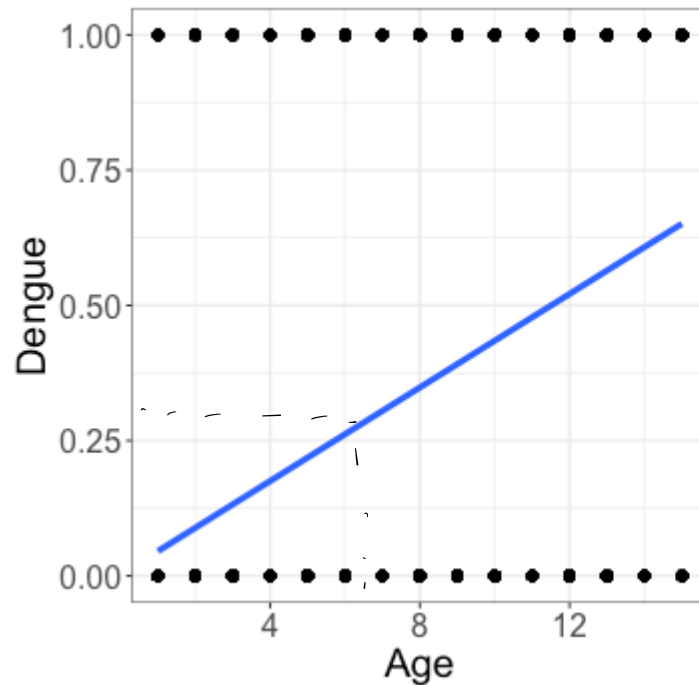
Warmup

$$\text{log-odds} = \log(\text{odds}) = \log\left(\frac{\pi}{1 - \pi}\right)$$

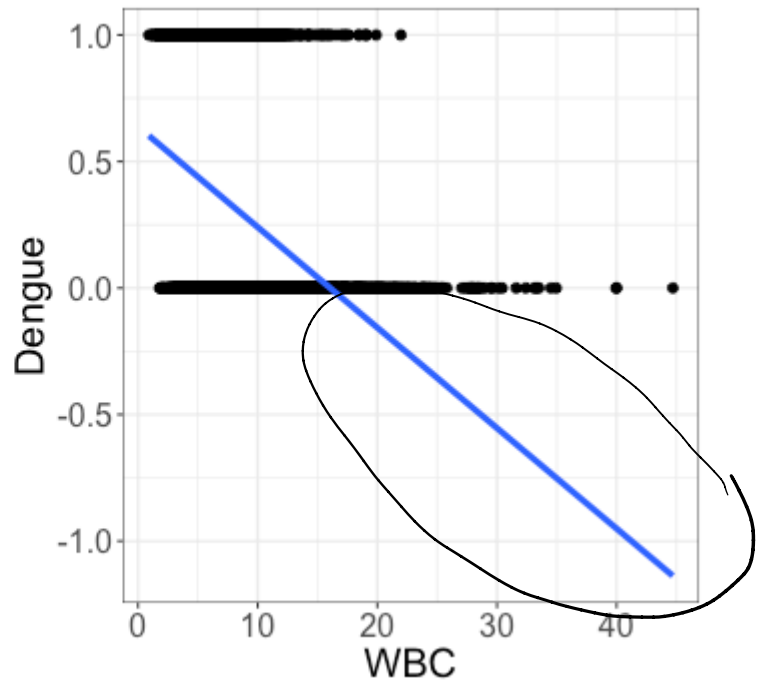


Last time

Don't fit linear regression with a binary response



$$\hat{\text{Dengue}} = 0.25 \quad ?$$

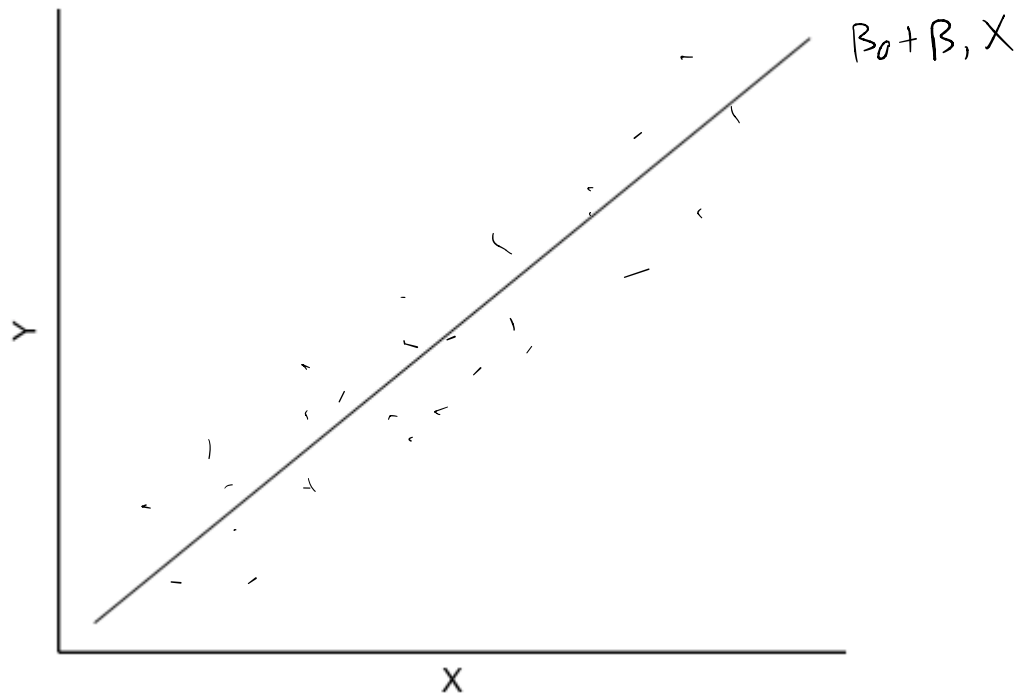


$$\hat{\text{Dengue}} < 0 \quad ??$$

Revisiting the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

↙ intercept ↘ slope



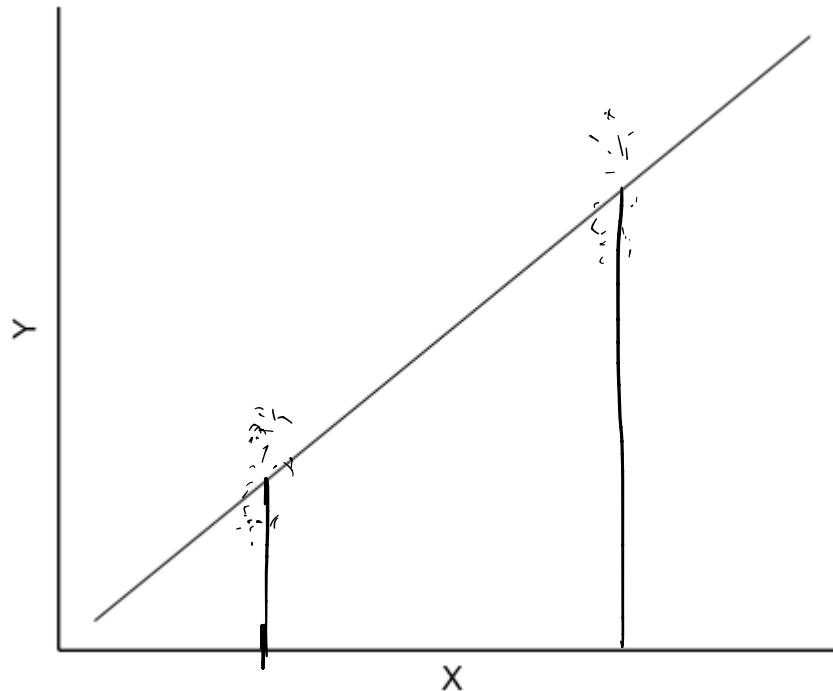
Will all of the observations fall exactly on the line?

No.

Revisiting the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$



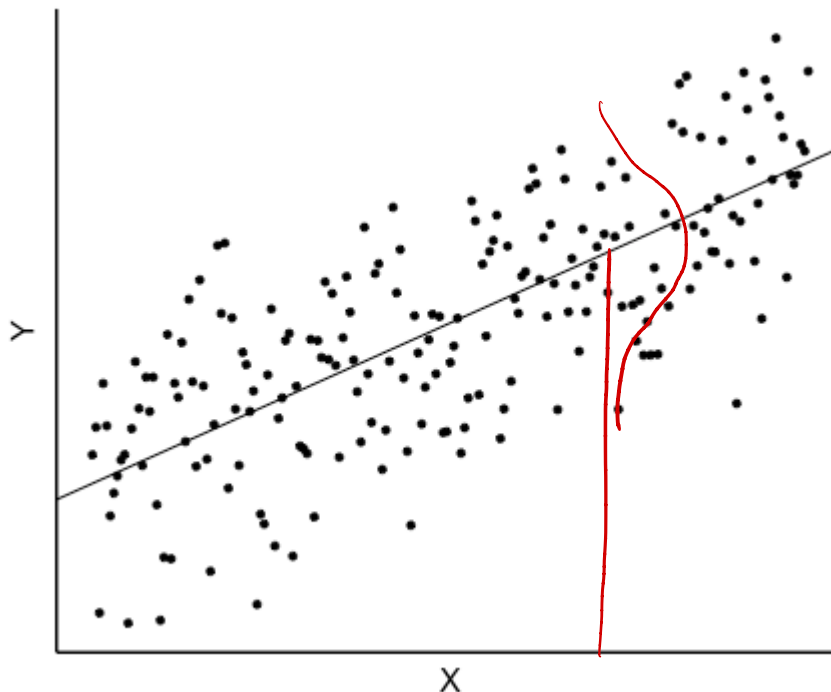
Given a value of X , how do I know where the values of Y are likely to be?

Revisiting the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

normal
errors



What do we often assume about the distribution of ε ?

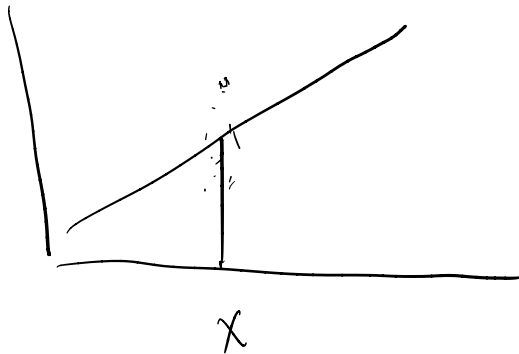
Revisiting the linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \varepsilon_i \sim N(0, \sigma^2)$$

$\Rightarrow y_i$ is also normal

$$y_i \sim N(\underbrace{\beta_0 + \beta_1 x_i}_{\text{line}}, \sigma^2)$$

↑
variability
around line



$$y_i \sim N(\overset{\text{mean}}{\mu_i}, \overset{\text{variance}}{\sigma^2})$$

← specifies
distribution
of y

$$\mu_i = \beta_0 + \beta_1 x_i \quad \leftarrow \begin{array}{l} \text{relates} \\ \text{distribution} \\ \text{to } x_i \end{array}$$

Parametric modeling

A regression model is an example of a more general process called **parametric modeling**

- + **Step 1:** Choose a reasonable distribution for Y_i e.g. $Y_i \sim N(\mu_i, \sigma^2)$
- + **Step 2:** Build a model for the parameters of interest
- + **Step 3:** Fit the model e.g. $\mu_i = \beta_0 + \beta_1 X_i$

e.g. in R: `lm(y ~ x, ...)`

Step 1: Choose a reasonable distribution for Y_i

What do I mean by a *distribution*?

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What do I mean by a *distribution*?

- + A **distribution** tells us what outcomes are possible for Y_i , and how often these outcomes occur.

Here the possible values of Y_i are 0 (no dengue) and 1 (dengue).

(binary
response)

How often do these values occur in the population?

Step 1: Choose a reasonable distribution for Y_i

What do I mean by a *distribution*?

- + A **distribution** tells us what outcomes are possible for Y_i , and how often these outcomes occur.

Here the possible values of Y_i are 0 (no dengue) and 1 (dengue).

How often do these values occur in the population?

- + We don't know, so we will estimate from the sample
- + We assume the probability $Y_i = 1$ depends on Age_i

Step 1: Choose a reasonable distribution for Y_i

How should I describe the distribution of Y_i ?

$$Y_i = 0 \quad \text{or} \quad 1$$

$$\pi_i = P(Y_i = 1)$$

$$\Rightarrow P(Y_i = 0) = 1 - \pi_i$$

That's it!

Bernoulli distribution

Definition: Let Y_i be a binary random variable, and $\pi_i = P(Y_i = 1)$. Then $Y_i \sim \text{Bernoulli}(\pi_i)$.
↑ "is distributed as"

What do I mean by a *random variable*?

Bernoulli distribution

Definition: Let Y_i be a binary random variable, and $\pi_i = P(Y_i = 1)$. Then $Y_i \sim \text{Bernoulli}(\pi_i)$.

What do I mean by a *random variable*?

A **random variable** is an event that has a set of possible outcomes, but we don't know which one will occur

- + Here $Y_i = 0$ or 1
- + Our goal is to use the observed data to estimate $\pi_i = P(Y_i = 1)$

Second attempt at a model

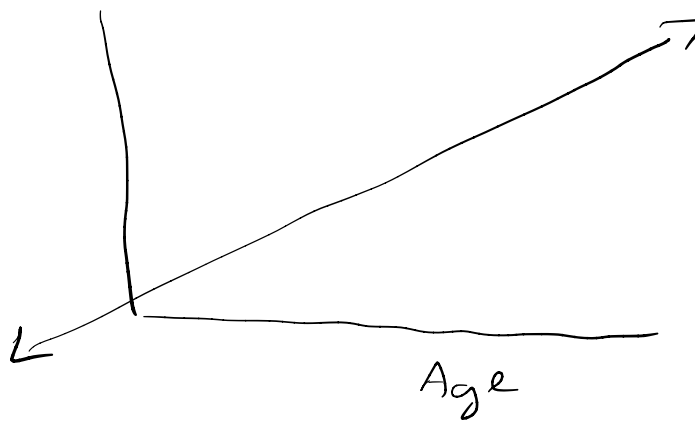
$$Y_i \sim \text{Bernoulli}(\pi_i) \quad \pi_i = P(Y_i = 1 | \text{Age}_i)$$

$$\pi_i = \beta_0 + \beta_1 \text{Age}_i$$

Are there still any potential issues with this approach?

$$\pi_i \in [0, 1]$$

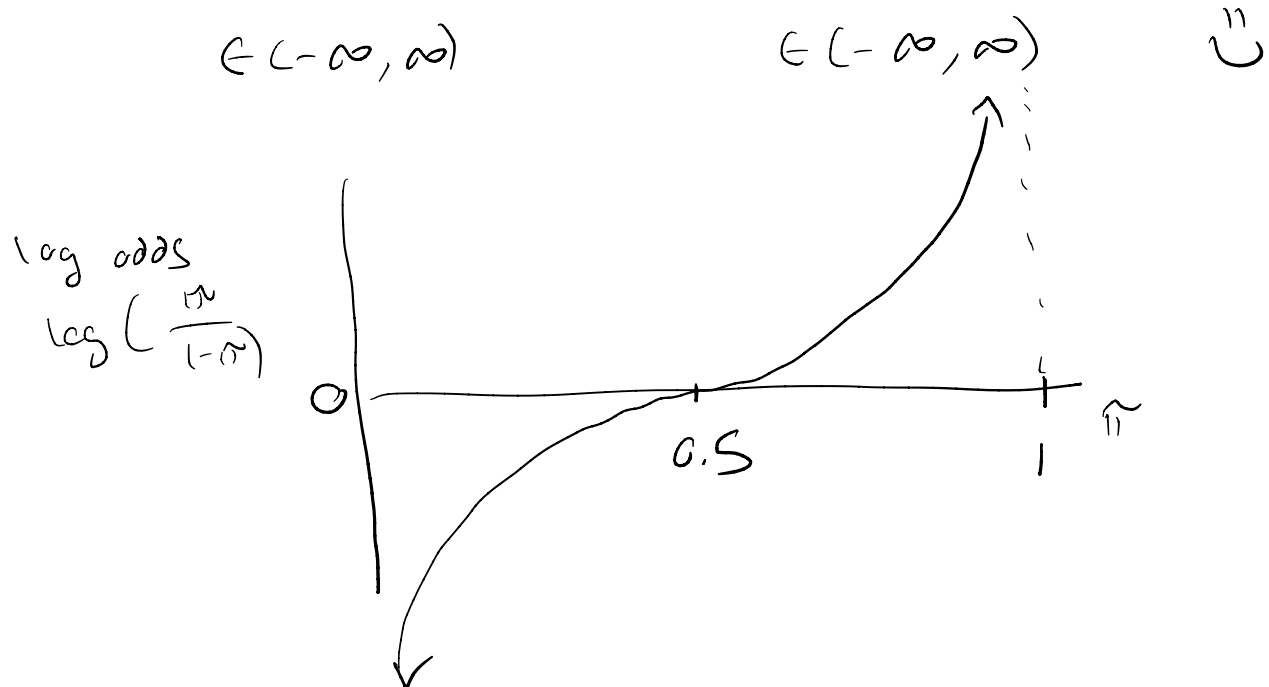
but $\beta_0 + \beta_1 \text{Age}_i$ does not
have to be banded



Fixing the issues

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\underbrace{\log\left(\frac{\pi_i}{1-\pi_i}\right)}_{\in (-\infty, \infty)} = \underbrace{\beta_0 + \beta_1 \text{Age}_i}_{\in (-\infty, \infty)}$$



Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

Random component: $Y_i \sim \text{Bernoulli}(\pi_i)$

specifies distribution of π_i

Systematic component: $\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 Age_i$

relates distribution

to explanatory variable(s)

Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

Random component: $Y_i \sim \text{Bernoulli}(\pi_i)$

Systematic component: $\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 Age_i$

Why is there no noise term ε_i in the logistic regression model?
Discuss for 1--2 minutes with your neighbor, then we will discuss as a class.

Fitting the logistic regression model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

```
m1 <- glm(Dengue ~ Age, data = dengue,  
          family = binomial)  
summary(m1)
```


Fitting the logistic regression model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

```
m1 <- glm(Dengue ~ Age, data = dengue,  
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summary(m1)
```

...

Coefficients:

##		Estimate	Std. Error	z value	Pr(> z)	
##	(Intercept)	-2.454345	0.075068	-32.70	<2e-16	***
##	Age	0.217312	0.008826	24.62	<2e-16	***

...

Class activity

- + Work with a neighbor on the class activity (handout)
- + I will collect your work at the end of class

For next time, read sections 6.4 and 6.6 in the textbook